01 — Project Proposal

Prepared 9 November 2026 for teacher review on 10 November. **Draft; not approved or implemented.**

Use the meeting proposal PDF as the presentation summary. The detailed sections below are supporting evidence. The three-sprint meeting schedule supersedes older planning proposals for discussion; approval is pending.

Source: 01-project-proposal.md

Project proposal

> **9 November planning update:** Latest user-provided site screenshots show three development sprints starting November 30, different task allocations, and release/presentation dates through February 3. Earlier dates/allocations below are historical proposals, not the current meeting commitment. Use the November 10 meeting package to review the new baseline and January 25 editing-cutoff question. Approval remains pending.

Version: 0.1 · **Prepared:** 9 November 2026 · **Status:** Proposed review draft; no teacher approval or product test execution claimed.

1. Proposed project

Develop ClassroomCast for local classroom screen/audio delivery, participation control and administrator-managed website restriction profiles. The expected educational value is readable demonstrations and observable, recoverable classroom operations during WAN outages. The client problem and benefits require teacher validation rather than unsupported claims of measured improvement.

2. Deliverables and boundaries

Deliver reviewed requirements, reusable local application/media/enforcement integration, both source deployment-mode demonstrations unless scope is formally revised, acceptance evidence, operating guides, source/demo links and a final academic report. Reuse an existing SFU; custom SFU construction, video archive, LMS/ERP integration and remote student-screen control are proposed exclusions.

3. Method and governance

Use six proposed two-week Scrum sprints within course milestones. Perform trusted-HTTPS, SFU and authorized enforcement feasibility work early. Demonstrate increments against traceable tests; teacher approval of release scope and campus IT permission are separate from team estimates. Exact stack remains provisional.

4. Resources and cost

Five named members have previously reported about 10 hours per week each. The prior 420-hour work estimate plus 42-hour contingency and 18-hour management reserve gives a provisional 480-hour envelope. The 600-hour nominal twelve-week capacity includes elapsed time and is not remaining capacity on 9 November. Hardware, certificate provisioning and cash budget are unknown; no procurement is authorized by this proposal.

5. Approval requested

Resolve target capacity, portable deployment inclusion, stakeholder authority, browser matrix, enforcement scope/failure rule, acceptance targets and current calendar. See teacher checklist. Approval name/date/evidence: **Pending**. No sponsor signature or funding commitment is inferred.

6. Supporting plan

Use scope, feasibility, schedule, risks, and the original PROJECT-PLAN.md for detailed WBS/effort provenance.

Source: 02-project-scope.md

Project scope

Version: 0.1 · **Prepared:** 9 November 2026 · **Status:** Proposed review draft; no teacher approval or product test execution claimed.

1. Release scope

Retain all supplied Must capabilities, the individually unlabeled campus/portable obligations and shared architectural requirements. Source scope includes local screen/audio broadcasting, class admission and control, policy administration/apply/verification/restoration, offline operation, diagnostics and authorized network setup. The teacher-specified capacity is 50–150 students. D12 concerns test conditions and evidence.

2. Demonstrable MVP versus full release

An early feasibility increment is one teacher and two laptops using trusted local access, valid joining and SFU playback, plus a separate authorized policy apply/restore spike. It is an early demonstration, not acceptance of the full required release. The second gate integrates permissions, supported audio, reconnection, policy state and safe recovery. Scaling and both modes must later be measured or explicitly changed through approval.

3. Lower-priority work

Should: richer diagnostics, materials, optional identification/downloads, scheduling/import-export/local list maintenance. Could: history, participation and supported group targeting. Global offline obligations remain required even though extra list-management convenience has a Should label. Optional features still require authorization and tests if delivered.

4. Proposed exclusions — needs agreement

ID	Proposed exclusion	Boundary
W01	Custom SFU implementation	Reuse a tested local SFU and integrate its controls
W02	Mandatory native student app	Preserve browser-only participation

ID	Proposed exclusion	Boundary
W03	Recording/archive of classroom video	History summaries do not imply recordings
W04	LMS/ERP integration	No integration requirement supplied
W05	Remote control or capture of student screens	Roster monitoring concerns participation/connectivity

5. Change control

For every requested scope change record source IDs, new wording, justification, dependency, affected tests, hours/cash/schedule impact and approving authority. Do not remove Must requirements merely to fit a sprint. See decision log.

Source: 03-project-objectives-and-expected-outcomes.md

Objectives and expected outcomes

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1. Measurable outcome candidates

Objective	Required evidence	Proposed measure / gate
Make teacher content available locally	F04/F05/F06 demonstration and TC-T01/S02/S03	Supported source displayed to admitted viewers
Preserve classroom control	Permission and admission tests	Denied cross-class/publish/rejoin attempts; TC-AUTH
Operate without WAN	Fresh-browser local test and dependency inventory	AT07; no required runtime cloud fetch
Apply only authorized restrictions	F10 read-back and affected-client evidence	Denied domain, allowed control, local classroom available
Recover temporary network changes safely	End/expiry/emergency/restart trials	AT05; visible unresolved result when unreachable
Support agreed scale and quality	Physical and simulated counts reported separately	AT02/03; capacity D12 pending
Enable teacher operation and handover	Observed tasks and runbook walkthrough	Prior proposed usability median 4/5; approval pending

2. Deliverable evidence register

Each outcome needs build/configuration, topology, actor/tester, timestamp, expected/actual result, evidence link, defects and reviewer. Current evidence status: **Not run / not collected**. A diagram or presentation is design evidence, not a successful software test.

3. Benefit evaluation

Compare observed task completion and reported legibility before and after a pilot if the teacher approves the method. Avoid causal claims from a small uncontrolled pilot. Record participant numbers and limitations in the final report.

Source: 04-feasibility-study.md

Feasibility study

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1. Current conclusion

Feasibility is **unproven**. The architecture is a plausible proposal based on local services and reuse, but no delivered benchmarks or approved campus integration establish success.

2. Feasibility experiments

Spike	Setup and procedure	Pass evidence	Failure response / owner
SP01 trusted offline access	Provision approved name/cert; fresh browsers on two laptops; remove WAN	Capture and app/media connections without trust bypass	Resolve trust with NET; Mengkong
SP02 SFU publication	One teacher, two viewers; screen and supported audio; zero-viewer start	Authorized publish/subscribe and observed media; supported matrix	Check transport/source compatibility; Dyhengmesa
SP03 enforcement	Isolated authorized scope; preserve baseline; apply test profile	Read-back plus denied/allowed/local-path controls; prior policy restored	Unsupported scope stays unavailable; Mengkong/Mesa
SP04 resilience	Interrupt backend/SFU/enforcement independently; replay pending operation	Truthful separate status, no stale admission or unsafe restore	Open defects and recovery decision; Sengleang/Hour
SP05 staged capacity	Document hardware/APs; increase subscriptions and physical clients	Resource/quality results at each agreed stage	Resolve D12 or equipment need; Mengkong

3. Operational and schedule feasibility

Confirm server access, Ubuntu version, resources, existing workloads, network owner, maintenance windows and pilot devices. Permission to host does not authorize DNS/firewall modification. Six-sprint dates are inherited planning dates; November 9 reforecast is required. Scope/hours cannot be validated from diagram completion alone.

4. Economic feasibility

Reuse existing proposed SFU/filtering components and inventory borrowed resources first. Costs and licenses for selected versions must be checked before adoption. Record quotes only if procurement is needed; do not present unknown expenses as zero. The 480-hour planning envelope is effort, not a monetary budget.

5. Gate result template

For each spike record hypothesis, authorization, configuration, observations, pass/fail against agreed criteria, limitations, decision and reviewer. All five spikes are **Not run in this package**. Proceed with full implementation commitments only after the relevant dependency is evidenced or explicitly accepted as a risk.

Source: 05-project-timeline-and-milestones.md

Timeline and milestones

> **9 November planning update:** Latest user-provided site screenshots show three development sprints starting November 30, different task allocations, and release/presentation dates through February 3. Earlier dates/allocations below are historical proposals, not the current meeting commitment. Use the November 10 meeting package to review the new baseline and January 25 editing-cutoff question. Approval remains pending.

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1. Planning basis

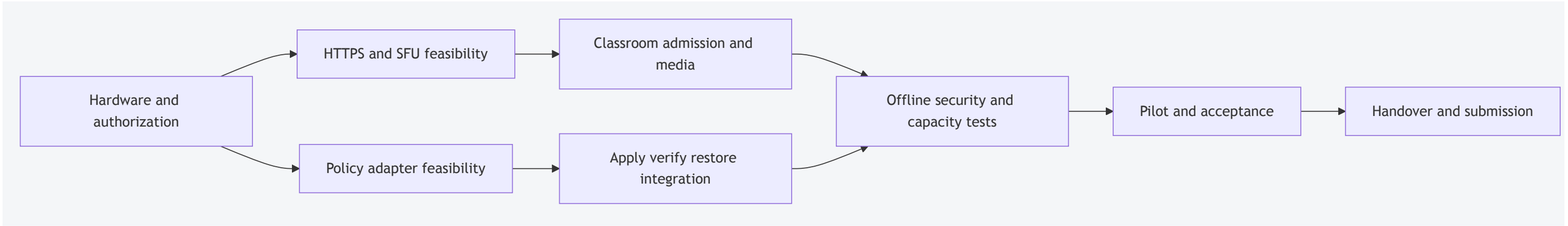
The following dates are retained from the prior plan, not claims of work completed. On 9 November 2026 Sprint 1 is already partly elapsed. Confirm actual work, remaining hours and calendars immediately; a newly written plan cannot retroactively supply November 2–8 capacity. Planning timezone: Asia/Phnom_Penh; verify course website deadline timezone.

Sprint	Dates	Goal / review evidence
1	2–15 Nov 2026	Feasibility and two-viewer local demonstration; G1
2	16–29 Nov	Class permissions, controls, audio and reconnect
3	30 Nov–13 Dec	Profiles, verified apply and restoration; G2
4	14–27 Dec	Remaining required behavior, failure recovery and both deployment modes
5	28 Dec–10 Jan 2027	Capacity/offline/security evidence; G3 release candidate
6	11–24 Jan	Pilot, corrections, acceptance, documentation and demo; G4 readiness Jan 22

Final course editing deadline: **25 January 2027, 23:59**. Use Jan 25 as submission contingency, not planned core implementation time. Proposed physical pilot Jan 11–15 requires booking and participants.

2. Dependencies

Diagram 1



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

3. Milestone exit criteria

G1: access/trust/media feasibility and unresolved scope recorded. G2: core classroom plus actual policy verification/restoration demonstrated. G3: required tests executed, defects triaged, physical/simulated scale limitations explicit. G4: acceptance decision, guides, reproducible demo and source links ready. Gates are **Pending**.

4. Forecast control

Track actual hours and remaining estimates by WBS; confirm individual calendars each sprint. Move lower-priority scope first; seek teacher decisions for Must scope/deadline trade-offs. No mathematical critical path is claimed without resource calendars and refined activity durations.

Source: 06-risks-and-mitigation.md

Risks and mitigation

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Version: 0.2 · **Prepared:** 9 November 2026 · **Status:** Proposed review draft; no teacher approval or product test execution claimed.

The team roster supplied by the user lists **HENG NGUON HOUR** (representative), **BUN SENGLEANG**, **CHANTHA MENGKONG**, **HAK SENGKEA**, and **HONG DYHENGMESA**. Full roster names are used below. Risk assignments remain proposed; this revision corrects naming without reallocating responsibilities.

1. Inherited risk register

Scores are provisional ordinal judgments (probability/impact 1–5), not measured probabilities. Review each sprint; score ≥12 or impact 5 merits active escalation. Historical response dates in the old plan need current follow-up.

ID	Cause → event → effect	P/I	Proposed risk owner	Prevention, trigger and fallback
R01	Campus control access unknown → integration unavailable → campus enforcement cannot be accepted	4/5	CHANTHA MENGKONG	Obtain IT decision in Sprint 1; trigger: no workable path by 13 Nov; use authorized portable network for development and escalate campus gap
R02	High viewer density → Wi-Fi saturation → freezes and capacity failure	4/5	CHANTHA MENGKONG	Wired server, staged tests and measured AP placement; trigger: loss/delay above targets at next load stage; lower agreed media profile/add APs or seek scope decision
R03	Laptop trust provisioning unknown → screen/SFU connection fails → offline demo blocked	3/5	CHANTHA MENGKONG	Two-laptop HTTPS test in Sprint 1; trigger: untrusted endpoint; obtain campus trusted cert/name or agreed trust provisioning and retest
R04	Policies span sessions/process restarts → incomplete restore → network remains restricted	3/5	HONG DYHENGMESA	Durable lease/snapshot, idempotent recovery worker and restore drill; trigger: rules differ after end/expiry; emergency restore with visible failure and operator runbook

ID	Cause → event → effect	P/I	Proposed risk owner	Prevention, trigger and fallback
R05	DNS bypass is possible → users reach restricted services → enforcement overstated	4/4	CHANTHA MENGKONG	Agree DNS scope/limitations; test alternate resolver/VPN cases on lab equipment; stronger authorized gateway/device controls or accepted limitation
R06	Audio varies by platform → expected source unavailable → media acceptance fails	3/3	CHANTHA MENGKONG	Publish compatibility matrix early; trigger: missing required audio source; agree supported combination and explicit fallback
R07	Initial work exceeds available hours → delivery slips → Must scope incomplete	4/5	HENG NGUON HOUR	Refine estimate and actual calendar by 15 Nov; trigger: forecast above remaining capacity; defer Should/Could and obtain explicit Must trade-off decision
R08	Privileged credentials or tokens mishandled → unauthorized control → demo/data compromised	2/5	HENG NGUON HOUR	Server-only secrets, scoped tokens and permission tests; trigger: exposed credential or bypass; revoke/rotate and stop affected admin action until corrected

2. Current additions

ID	Cause / event / consequence	Proposed risk owner	Response and trigger
R09	Insufficient physical devices or Wi-Fi evidence prevents demonstration of the specified 50–150 capacity	HENG NGUON HOUR	Agree D12 test conditions; plan 150-student capacity evidence and distinguish physical from simulated clients
R10	School host chosen without inventory; deployment may conflict with existing workloads	CHANTHA MENGKONG	Inventory and maintenance authorization before installation
R11	Repeated generated diagrams diverge from code; team implements wrong paths	BUN SENGLEANG	Review Mermaid as source; deterministic rendering; trace every failure outcome
R12	Pending semantic decisions accumulate; late rework delays release	HENG NGUON HOUR / HONG DYHENGMESA	Assign D01–D20 owners; prioritize admission, authority and restoration before sprint commitment

3. Issue and escalation template

Record ID, opened date, trigger observed, impact, owner, next action, due date, decision needed and evidence. An observed problem becomes an issue; do not keep calling an active failure only a risk. Network impact escalates promptly to the authorized operator. Other external blockers go through HENG NGUON HOUR under the prior one-working-day escalation proposal.

4. Residual risk acceptance

Teacher and network owner must explicitly accept limitations relevant to their authority. A portable demonstration does not prove campus integration. No known risk has been recorded as accepted or closed by this package.